

CHE 2201 – Physical Chemistry II

Phase Equilibria

Lec. 3. Two Component (Binary) Systems. Raoult's Law

3.1. Introduction

As an example of a two component system, we will study a liquid solution, which is formed by adding another pure liquid to a one component system. Then the behaviour of the system changes dramatically with the proportions of both components. From experiments, it has been observed that the liquid mixture which boils over a range of T_s , has a vapour composition different to that of the liquid mixture.

Minimum number of phases possible for a two component system is one ($P = 1$), i.e. either vapour, liquid or solid phase. Thus, the application of the Phase Rule for a two component system gives us:

$$F = C + 2 - P \quad (3.1)$$

$$F = 2 + 2 - 1 = 3.$$

$F = 3$ means that we need three independent variables to define the state of a two component system. We have already studied two such variables under one component system, i.e. T and p . As such, in order to explain the phase equilibria of a two component system, it is necessary to introduce another variable that is independent of other two. The obvious variable is the amount of one component added to a known amount of other component. This can be interpreted as the amount of one component relative to the amount of the other component, or in other words, we can introduce it as the mole fraction.

The mole fraction of a liquid A (X_A) in a mixture of liquids A and B can be written as

$$X_A = n_A / n_{\text{Total}} = n_A / (n_B + n_A) \quad (3.2)$$

where n_A and n_B are the number of moles of A and B present in the mixture.

It can be proved that

$$X_A + X_B = 1 \quad (3.3),$$

or in other words, the sum of mole fractions of two components in a mixture is equal to one, and it is true for any number of components.

Thus, in order to represent the behaviour of a two component system, we need to study the variation of three variables, i.e. T, p and mole fraction. As mentioned earlier (Refer to section 2.3), the 3-D phase diagrams are very inconvenient in use. Therefore, for convenience, most experiments and industrial processes are conducted in such a way that one of the variables is kept constant. Then a graph can be drawn using two variables, keeping the other variable constant. Three plots can be constructed this way.

1. Pressure vs. Mole Fraction at constant temperature
2. Temperature vs. Mole Fraction at constant pressure
3. Pressure vs. Temperature at constant Mole Fraction

From these graphs, we would be able to understand that the distillation of a mixture of liquids could be carried out at const. p as well as const. T. In general, most distillations (e.g. petroleum oils) are carried out at const. p. (We will discuss the distillation processes in detail, during next lecture).

3.2. Raoult's Law and the Ideal Solutions

Raoult's Law states that, "*the vapour pressure of a liquid in a liquid mixture depends on the mole fraction of the liquid, and not on the nature of the other component present.*"

Hence, we can write

$$p_A \propto X_A \quad \text{or}$$

$$p_A = K_A X_A \quad (3.4)$$

where p_A is the vapour pressure; X_A is the mole fraction of liquid A; and K_A is a constant.

On the other hand, for a pure liquid: $X_A = 1$.

It follows that:

$$p_A^0 = K_A \quad (3.5),$$

where p_A^0 is the vapour pressure of the pure liquid.

From equations (3.4) and (3.5), we have

$$p_A = p_A^0 X_A \quad (3.6)$$

The equation 3.6 can be used to calculate the partial pressure of component A in a binary system having a known mole fraction of X_A .

Solutions obeying the Raoult's law in the form of equation 3.6, at all concentrations and all T s are called ideal solutions. However, only a few solutions obey this law with sufficient accuracy at any concentration, i.e. at values of X changing from 0 to 1. They are the limiting, simplest kind of liquid solutions. For example, benzene and toluene form an ideal solution.

This means that the forces between molecules of one component in the solution, do not affect the forces between the molecules of the other component. In other words, if the two components are A and B, the forces between A and B molecules may be identical with those between A and A, and between B and B. As such, when two components are mixed to form an ideal solution, no heat is evolved or absorbed during the process. If both components are liquids, then total volume of the solution is equal to the sum of the volumes of each component taken separately.

For an ideal solution of two components A and B, according to the Raoult's law, we can write:

$$p_A = p_A^0 X_A \quad \text{and} \quad p_B = p_B^0 X_B \quad (3.7),$$

where X_A and X_B are the mole fractions of A and B in the mixture;

p_A^0 and p_B^0 are the vapour pressures of A and B in the pure state.

The total vapour pressure of the solution can be obtained by using the Dalton's law of partial pressures.

Dalton's law of partial pressures states that, "*the total pressure exerted by a mixture of gases behaving ideally, is the sum of the pressures exerted by individual gases occupying the same volume alone*".

Thus, at any value of mole fraction, the total vapour pressure (p) is given by:

$$p = p_A + p_B \quad (3.8)$$

From equations (3.7) and (3.8), we get

$$p = p_A^0 X_A + p_B^0 X_B \quad (3.9)$$

On the other hand

$$X_A + X_B = 1 \quad (3.3)$$

$$X_A = 1 - X_B \quad (3.10)$$

From equations (3.9) and (3.10), we get

$$p = p_A^0 (1 - X_B) + p_B^0 X_B$$

$$p = (p_B^0 - p_A^0) X_B + p_A^0 \quad (3.11)$$

[Equation 3.11 is of the type $y = mx + C$]

Plots of partial pressures for various X of components A and B in the liquid phase are shown in the Fig. 3.1, along with the plot of total vapour pressure versus X. Since the partial pressure of each component is directly proportional to its mole fraction (See equation 3.7), a plot of partial pressure versus X should be a straight line. Similarly, the graph showing the total pressure is also a straight line (See equation 3.11).

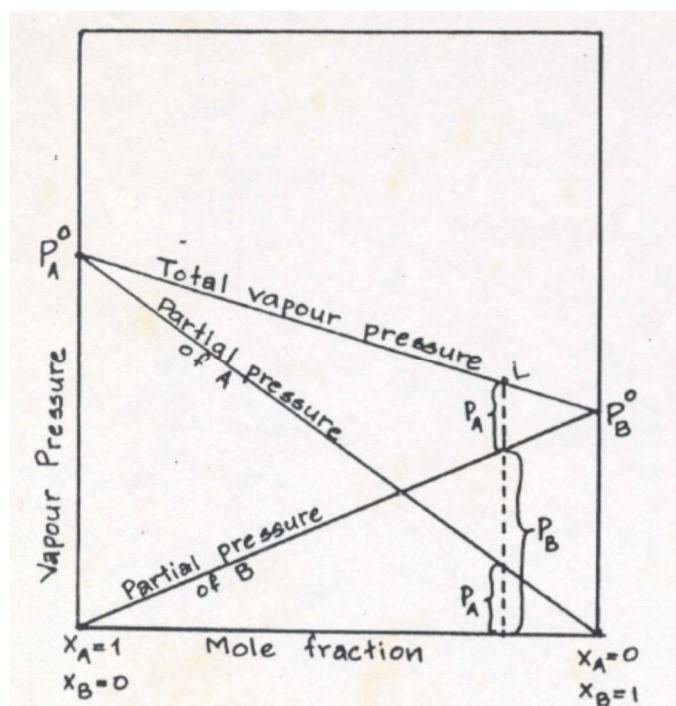


Fig. 3.1. Vapour Pressure of an Ideal Solution at const. T

The total pressure curve ($p_A^0 p_B^0$) is called the liquid composition curve. Any point on this line indicates the ratio of X_s in the liquid mixture at that pressure. Pressure at any point on the line $p_A^0 p_B^0$ is equal to the sum of the partial pressures of two components (equation 3.8). For example, pressure p at point L on the line $p_A^0 p_B^0$ is equal to the sum of the two points p_A and p_B on individual lines.

Similar types of graphs can be obtained for completely miscible liquids of different boiling points, e.g. benzene and toluene. Usually in miscible volatile liquids, the composition of the vapour is different from the composition of the liquid, with which it is in equilibrium. It is obvious that the vapour phase should always be richer in more volatile component when compared to the liquid phase. In the case of the benzene/toluene mixture, vapour is richer in benzene than in the liquid phase, as benzene is more volatile than toluene.

In the Fig. 3.1, of the two components A and B , we have considered A to be more volatile than B ($p_A^0 > p_B^0$). Therefore, at a given pressure, the vapour phase consists of more A than B .

3.3. Real Solutions

Raoult's law is not obeyed by real solutions (i.e. non-ideal solutions). The vapour pressures of these solutions are greater or smaller than that of the ideal solutions. Non-ideal behaviour of solutions can be explained as due to the interaction (attraction or repulsion) of molecules present in the system. If the two components A and B in a system are different in character, the molecular forces between A molecules will differ from those acting between the B molecules. It means that the presence of B will affect the escaping tendency of A molecules and vice versa. As a result, the Raoult's law would not be applicable.

Two types of deviations from the Raoult's law are possible for real solutions. They are

1. Positive deviation from the Raoult's law, and
2. Negative deviation from the Raoult's law.

3.3.1. Positive Deviations from the Raoult's Law

If the two components differ sufficiently in polarity, then the repulsion between molecules will affect the vapour pressure of the resulting mixture. If the attraction between B molecules is much stronger than that between A molecules, then the effect is to force the A molecules out of the liquid into the vapour. In other words, the escaping tendency of A molecules will be increased by the presence of B molecules. As a result, the partial vapour pressure of A will be greater than that is to be expected from the Raoult's law. Such a behaviour is known as the positive deviation from the Raoult's law.

When one component shows + deviation, then the other component also shows a similar deviation. As a result, the whole system shows a + deviation. In such systems, total vapour pressure will be greater than that calculated from the Raoult's law.

The extent of the deviation depends on the polarity of the two components. If the two liquids in the mixture are both non-polar (e.g. CCl_4 /heptane), or if both are moderately polar (e.g. ether/acetone), then the + deviations from the ideal behaviour are not large (See Fig. 3.2).

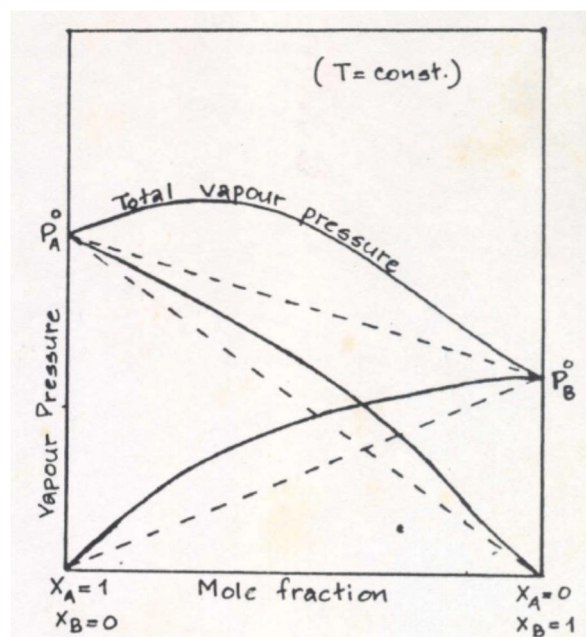


Fig. 3.2. Positive Deviations from Raoult's Law. Non-polar & Moderately Polar Systems

On the other hand, if one component is slightly polar and the other is highly polar, or if the mixture consists of a polar and non-polar compound (e.g. alcohol/hydrocarbon), considerable + deviation may occur. Such large + deviations may, sometimes lead to the liquids becoming partially immiscible (See Fig. 3.3).

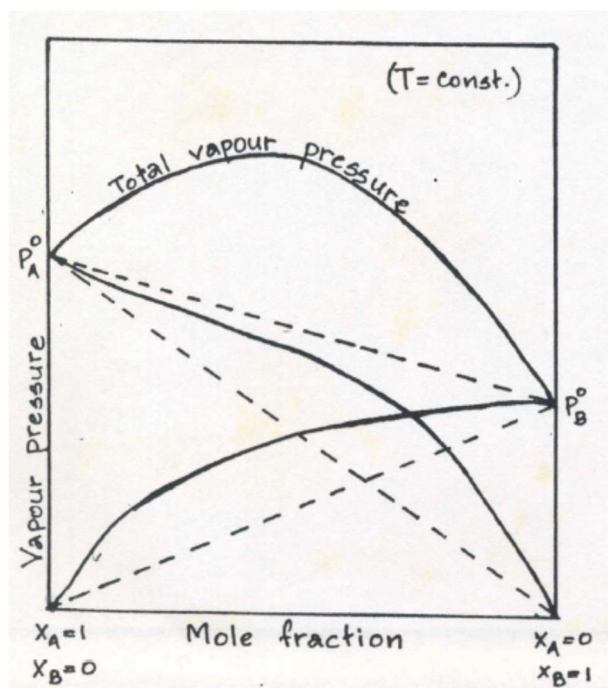


Fig. 3.3. Large Positive Deviations from Raoult's Law. Polar Systems

3.3.2. Negative Deviations from the Raoult's Law

If the molecules of the two components A and B of a mixture attract each other strongly, then the vapour pressure of each component will be less than that calculated from the Raoult's law. Such type of non-ideal behaviour is known as the negative deviation from the Raoult's law. (See Fig. 3.4).

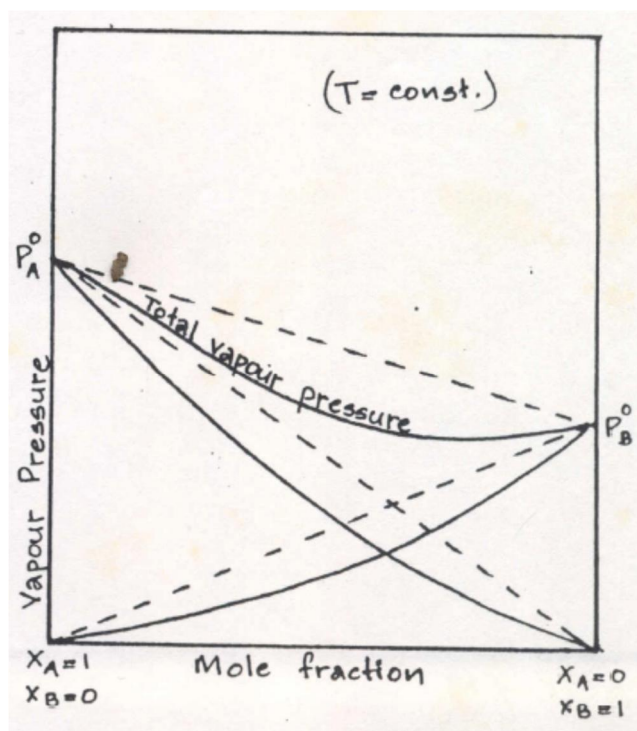


Fig. 3.4. Negative Deviations from Raoult's Law

Examples:

Pyridine/Acetic Acid;

Acetone/Chloroform;

Ethyl Ether/Chloroform;

Acetone/ CCl_4